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Joint resource allocation and congestion-aware routing based on hybrid optimization in IoT

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ABSTRACT

Presently, the Internet of Things (IoT) is linked through various forms of communication medium. In IoT, numerous devices are interconnected; hence, network congestion is created. The network congestion and overflow may create data privacy issues. Furthermore, resource allocation (RA) is the major function in IoT systems, in which the system's performance is based on the precise and effectual allocation of resources. As a result, this research proposes a novel optimization named Lyrebird Jaya Hunger Search Optimization (LJHSO) for RA and Lyrebird Fractional Walruses Optimization Algorithm (LFWBO) for congestion-aware routing in IoT. The IoT nodes simulation is the initial process and the cross-layer optimization approach for joint RA and routing is performed. The IoT layer is composed of sensing, network, data processing, and application layers. The RA is carried out in the physical or sensing layer using the LJHSO. Moreover, the congestion-aware routing is done in the network layer by the LFWBO with fitness parameters like queue length, energy, distance, and link quality. Furthermore, metrics like energy, makespan, throughput, and execution time are considered to validate the performance of the proposed LJHSO+LFWBO-based joint RA and congestion-aware routing, with the finest outcome of 0.747 bits/Joule, 0.332, 0.853 Mbps, and 1.913 s are attained.

1. Introduction

IoT often is termed as the Internet of Everything, which is employed as a new technological model for making envisions in the global network of machines and gadgets for performing effectual communication. The IoT is considered one of the crucial components in future technology and provides considerable attention from various sectors [1]. Moreover, the IoT is created by the merging of numerous operations like sensing, identification, computation, and networking. It permits value-added services and big-scale technical developments for personalizing the consumer's interactions with various "things". Several IoT applications are employed, which are categorized into various classes like traffic, health, retail, logistics, agriculture, smart metering, smart cities, process automation, and remote monitoring [2]. The rapid growth of IoT devices offered an unprecedented count of connected physical items. The linking of these IoT devices creates a large volume of data, which is necessitated for allocating a massive computer resource, communication bandwidth for analyzing the data, and storage space [3]. Moreover, the IoT model contains web-enabled smart devices, which include integrated systems like a Central Processing Unit (CPU), communication gear, and sensors for collecting and sending the data [4]. IoT has provided many advantages owing to its broad range of usages in biomedical, industrial, smart cities, environmental monitoring, and agriculture. Hence, the IoT is employed as the internetworking of physical things in the day-to-day function of real-world scenarios, and it provides the standard communication framework to offer more services to end users [5].

In IoT, the linked devices are considered objects in the Internet platform. These devices are employed for assigning and sending high-quality resources to the users [6]. RA is considered an important function in IoT because it offers reliable resource distribution and manages the Quality of Service (QoS). Based on the criteria, the data streams are grouped, and the different kinds of network attacks are mitigated to provide a high level of security. Moreover, computing power, storage capacity, and exchangeable energy are considered accessible resources. Owing to the effectual allocation of IoT resources, the network's performance has enhanced. IoT is a heterogeneous, and distributed network; hence, proper RA is needed [7]. For RA, Machine learning (ML) [28] techniques are employed, which depend on computational intelligence like reinforcement and supervised learning. Deep reinforcement learning (DRL) algorithms are also utilized for optimizing the resources

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in numerous computing paradigms [6]. Due to the variations of physical load on the server, the RAs are controlled proactively. Dynamic RA is more complex, especially when the QoS requirements are changed over time, minimizing the processor stability period, and computing the processor availability [8]. In addition, RA performs firmware upgrades, network factors, performance computation, and so on. The RA interfaces are commonly set at one end of the network [29] and provide the related data, which the device controller considers a massive amount of data and components [6].

The major concern in the IoT network is energy constraint, which often controls the whole lifetime of the network. The network sensors have the potential of continuously gathering and monitoring the data, transmitting the data among the IoT networks, and providing long-term network communication [32]. One of the most essential tasks in IoT is energy-effectual routing, which minimizes energy and computational expenses. In IoT, low-power devices have been used to meet the criteria of node position via transmission and compilation. Routing protocols not only transmit the data; they also provide the extended lifetime of the networks [9]. Conversely, the unpredictable low energy and restricted resources typically require the provisioning of QoS, and it makes the individuality of IoT more complex to attain effective communication. Network congestion is the erratic behavior or unreliability of the network during high-traffic volume conditions. Therefore, nearby nodes can aggregate their energy during the routing, and the QoS criteria are considered for diminishing the overhead and ensuring better service quality via the data provisioning and data gathering process [10]. Congestion control is essential for maintaining the network stability and quality of experience (QoE) in the user to provide excellent QoS delivery [11]. Combinatorial optimization is employed to describe the issue of optimizing network parameters for extending the network lifetime [12]. Moreover, the meta-heuristic optimization approaches offered better results for routing [6].

Resource allocation and congestion-aware routing face key challenges such as balancing efficiency with fairness, handling dynamic demand, and managing limited resources. Both require real-time monitoring to adapt to fluctuating conditions while avoiding system overload and congestion. Ensuring scalability in large systems and minimizing overhead in decision-making processes add to the complexity. Additionally, optimizing for multiple performance metrics, such as latency and throughput, is difficult. Effective strategies must address these issues to maintain optimal system performance. Efficient resource allocation and congestion-aware routing are important for optimizing system performance, ensuring fairness, and minimizing delays. Addressing these challenges enhances scalability, reliability, and responsiveness in complex, dynamic environments. Thus, the main purpose of the work is to devise joint RA and congestion-aware routing in IoT using hybrid optimization. At first, the IoT nodes are simulated, and then a cross-layer optimization approach for joint RA and routing is carried out. The RA is done in the physical or sensing layer, where the LJHSO is employed. The congestion-aware routing is performed in the network layer using the proposed LFWBO with fitness parameters, like queue length, link quality, distance, and energy.

The remaining section of the paper covers six sections. The motivation, and review of prior works regarding the RA and congestion-aware routing in IoT is given in Section 2. Section 3 shows the system model of IoT, and Section 4 explains the LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT. Section 5 portrays the experimental result, and the conclusion is given in Section 6.

The major contribution of the paper is given below:

- **Proposed LJSHO-based RA:** In the RA of IoT, the LJSHO is employed. Here, the merging of the Lyrebird Optimization Algorithm (LOA) with the Jaya algorithm and Hunger Game Search Optimization (HGSO) develops the LJSHO.
- **Proposed LFWBO-based congestion-aware routing:** The routing of IoT is carried out using the proposed LJWBO. Here, the integration of LOA with Fractional Calculus (FC) concept, Walruses Optimization Algorithm (WaOA), and Dung Beetle Optimizer (DBO) develops the LFWBO.

2. Motivation

The effectual RA of IoT is needed to enhance the network's performance and extend the device's lifespan. In IoT, several devices are communicating concurrently; hence the congestion has occurred to create packet loss, degrade the performance, and increase the latency. Therefore, optimization-based RA and congestion-aware routing are developed.

2.1. Literature survey

Costa et al. [8] developed the Dijkstra algorithm for allocating resources in IoT devices. It provided effectual energy utilization and eliminated the blocking of data at the receiver. Still, it failed to explore the merits of varying the mono-objective issues for reducing the difficulties and independence among the objectives. Shuaib et al. [13] developed the Dynamic Energy-Efficient Load Balancing (DEELB) in IoT. It was a successful RA process with lesser packet loss. Still, it failed to examine the unfavorable service migration in numerous factors like migration cost, service migration, and data transmission. Lin et al. [14] devised the AI-driven collaborative dynamic RA (ACDRA) in IoT. This model minimizes the duration for making decisions and remains constant when the tasks are changed. However, it did not consider the joint control of parallel implementation of numerous tasks and the prevention of resource variance to execute numerous tasks. Ahmed et al. [15] developed the DL model for allocating resources in IoT. This model was used to perform the relay selection and optimal power allocation. However, it failed to consider the ML with blockchain for employing various applications. Adil [9] developed the dynamic hop selection static routing protocol (DHSSRP) in an IoT. Here, the DHSSRP was based on the priority-enabled data for managing the network, and it enhanced the lifespan of the devices. Still, the real-time usage of the model was poor. Praveen and Prathap [16] devised the energy-efficient congestion-aware RA and routing protocol (ECRR) for the IoT model. The packet loss, energy consumption, and delay of the model were minimal. In this model, the hybrid optimization approach generates the system complexity.

Thenmozhi et al. [10] developed the Energy-Efficient Priority-based Multi-Objective QoS routing (PMQoSR). In this model, the routing mechanism was used for selecting the congestion-aware route and providing the effectual utilization of resources. Still, it failed to provide security under the attack model of IoT cloud systems. Vankdothu and Abdul et al. [11] devised the Adaptive Krill Herd (AKH) optimization for congestion-aware routing and sustained the network to increase the survivability of the system. Still, the performance of the model depends on the traffic patterns, deployment scenarios, and network topologies.

Selvan [30] introduced Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) with multipath routing protocols for enhanced load balancing in data-heavy networks. Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) with multipath routing protocols were adaptable to dynamic changes in the network and had global search capabilities, but scalability was large, and implementation was high. Abdullah et al. [31] introduced Outer Approximation Algorithm (OAA) for navigating resource allocation and energy efficiency. OAA handled nonlinearities and had feasible convergence, but OAA had high computational issues.

The comparisons of existing methodologies are given in Table 1.

2.2. Challenges

The following challenges are faced by existing works regarding RA and congestion-aware routing:

Table 1
Comparison table for literature survey.

Sl. No	Author	Methodology	Advantage	Drawback
1.	Costa et al. [8]	Hybrid topology for IoT applications that combines mesh and star configurations	Energy efficiency, Scalability	High complexity, failed to reduce difficulties and dependencies among multiple objectives.
2.	Shuaib et al. [13]	Dynamic Energy-Efficient Load Balancing (DEELB) method	Optimal load distribution, Successful resource allocation process with reduced packet loss.	Did not examine unfavorable service migration factors such as migration cost, service migration, and data transmission.
3.	Lin et al. [14]	AI-Driven Collaborative Dynamic Resource Allocation (ACDRA) Algorithm	Optimal resource allocation	Failed to prevent resource variance for executing multiple tasks.
4.	Ahmed et al. [15]	Resource utilization and service delivery in IoT-based Wireless Sensor Networks (WSNs) using Deep Neural Network	High throughput, and High energy efficiency	Did not incorporate ML with blockchain for various applications.
5.	Adil [9]	Dynamic Hop Selection Static Routing Protocol (DHSSRP)	Effective load balancing	Complexity in network management
6.	Praveen and Prathap [16]	Energy efficient, congestion-aware resource allocation and routing protocol (ECRR) for IoT network based on hybrid optimization techniques	Less Energy consumption	Hybrid optimization approach increases system complexity.
7.	Thenmozhi et al. [10]	Energy-Efficient Priority-based Multi-Objective QoS routing (PMQoSR) mechanism using hybrid optimization algorithm	Effective resource allocation	Failed to provide security under IoT cloud system attack models.
8.	Vankdothu and Abdul et al. [11]	Congestion and Interference Aware Safe Routing Protocol	Utilizes adaptive fuzzy clustering, adaptive quantum logic-based encryption, and adaptive optimization methods (AKH) to dynamically adjust to varying network conditions, ensuring efficient routing.	High computational overhead, less scalability
9.	Selvan [30]	Genetic Algorithms and Particle Swarm Optimization (GA+PSO) with multipath routing protocols	Adaptability, Reliability	Computational complexity and premature convergence
10.	Abdullah et al., [31]	Outer Approximation Algorithm (OAA)	Handles multiple constraints	High computational cost

- In [8], the Dijkstra-based RA offered topological planning and provided efficient energy utilization. Nevertheless, it failed to enhance the delineation of RAin LoRa and ZigBee networks.
- The DEELB in [13] was effective for fog load balancing. Still, this model was failed to applicable in security concerns for orchestration, and service mitigation for dynamic load balancing in numerous conditions.
- The DHSSRP in [9] was used to maintain multiple routes and dynamically match traffic across the routes. Still, this model suffered due to the additional overhead related to memory usage, and computational resources.
- In [10], the PMQoSR was reliable in changing the network condition. In case, the protocol relies on pre-defined optimizing parameters, it may struggle to face the real-time changes in the traffic patterns.
- IoT networks contain a variety of devices with different capabilities, communication protocols, and resource constraints. Still, the monitoring of these heterogeneous devices was more complex due to the network traffic.

3. System model for IoT

IoT devices consist of four primary parts such as application, data processing, network, and sensor layers. The major function of the sensing layer is to find any phenomenon in the peripherals of the devices and gather real-time data. Fig. 1 exhibits the system model for IoT in RA and routing [17]. The sensor layers are used to allocate the resources effectively. Here, the LJHSO is employed for RA. This layer includes numerous sensors, which are the major aspects of IoT. The sensor hubs merge the sensors in IoT devices. Furthermore, the sensor hub acts as a common link for numerous sensors, which gather and send the sensor data to the processing device. The sensors in IoT devices are classified into three types such as environmental sensors, position sensors, and motion sensors. The pressure sensors and light sensors are considered the environmental sensors. The main function of the environmental sensor is to enable the devices to make independent decisions related to the variations in a device's periphery. The position sensors in the IoT device are used for monitoring the physical location of the device. The Global Positioning System (GPS), and magnetic sensors are the popularly utilized position sensors, in which the magnetic sensors are used as a digital compass for managing the alignment of a gadget display, and the GPS is used to navigate the IoT devices. Motion sensors are used for measuring the variations of direction and motion in the devices. Linear and angular motion are the two kinds of motion sensors, in which linear motion specifies the linear displacement, and angular motion represents the rotating displacement of the device. The network layer is employed as the communication channel to send the data from the sensing layer to other devices. It effectually performs the congestion-aware routing. Here, the routing is performed by the proposed LFWBO. In the network layer, communication technologies like Bluetooth, Wi-Fi, Zigbee, LoRa, Z-Wave, and cellular networks are used to allow data flow among the devices in a similar network. Furthermore, the data processing layer covers the central data processing unit and gathers the data for making decisions. Numerous IoT devices like smart home hubs and smartphones are presented in the data processing layer to enhance the user experience. This layer interacts with other connected devices through the network layer. The application layer is used for displaying the outcome of the data processing layer for completing the numerous applications of IoT devices. Moreover, the application layer acts as a user-centric layer to carry out several functions for users. The various IoT applications like smart homes, smart transportation, and healthcare are included in the application layer.

4. Proposed Jaya-HGS for optimal RA in IoT

In IoT, numerous devices are interconnected, which creates network congestion. Network congestion may affect data security. Furthermore, RA is the main issue in IoT, because the efficacy of the performance is based on the effectual allocation of resources. As a result, a novel optimization is developed for joint RA and congestion-aware routing. Initially, the IoT nodes are simulated, and then a cross-layer optimization approach for joint RA and routing is performed. In the physical or sensing layer, RA is accomplished by utilizing the LJHSO. On the other

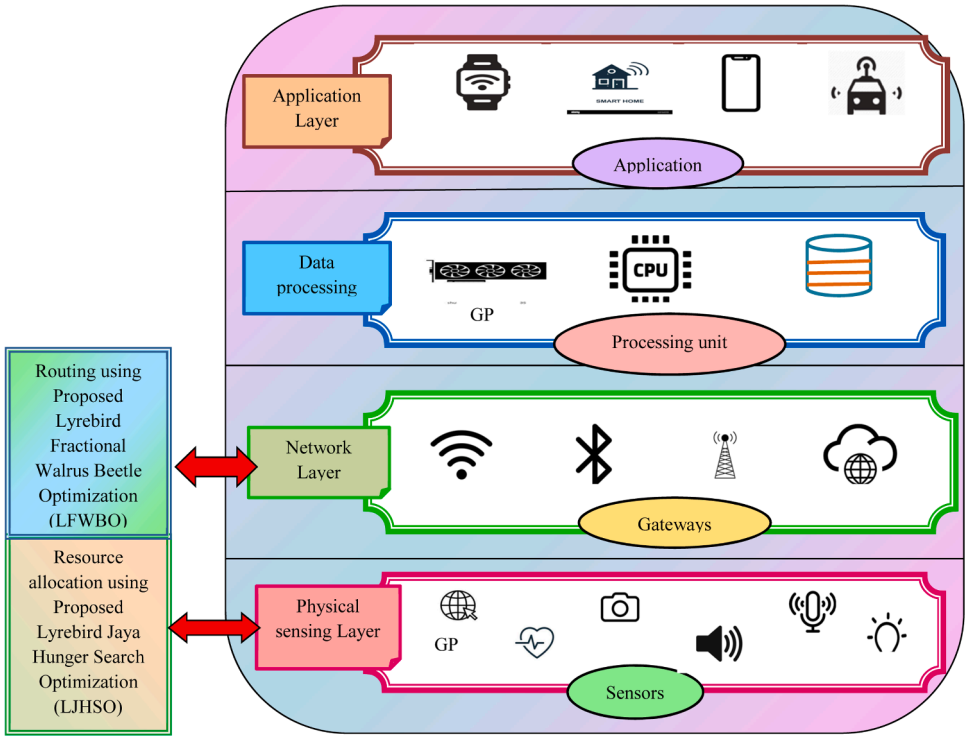

Fig. 1. System model for RA and congestion-aware routing in IoT.

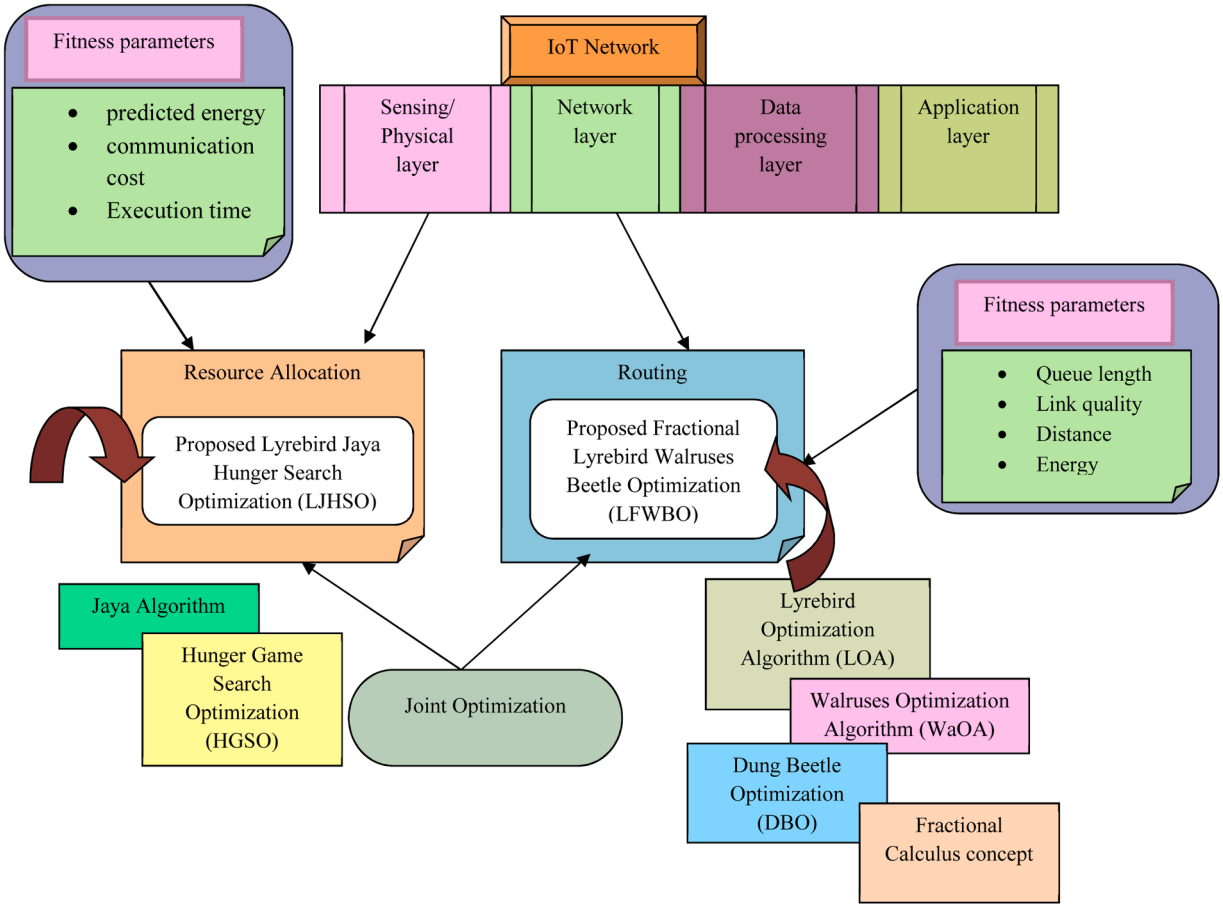

Fig. 2. Block diagram of LJHSO-based RA and LFWBO-based congestion-aware routing in IoT.

hand, congestion-aware routing is done by the LFWBO with fitness parameters, like queue length, link quality, distance, and energy. Fig. 2 shows the block diagram of Joint RA and congestion-aware routing in IoT.

4.1. RA at physical or sensing layer

The RA is carried out to allocate the resources effectually. Here, the RA is carried out using the LJHSO with objective functions such as predicted energy, communication cost, and execution time.

4.1.1. Multi-objective function

The multi-objective function is used to predict the energy for allocating the resources to the finest gateway; hence, the performance of the network is improved. Here, Deep LSTM is utilized to predict the energy, in which the input energy EN is used to predict the energyEN_{pr}

(a) Architecture of Deep LSTM

LSTM [18] is an advanced version of Recurrent Neural Networks (RNNs) designed to overcome the limitations of traditional RNNs, particularly their inability to handle long-term dependencies in sequences. Using LSTM enables efficient prediction of network congestion patterns and dynamic resource allocation, improving network performance and reducing delays in real-time IoT applications. This approach enhances the scalability and adaptability of IoT networks under varying traffic conditions. LSTMs learn from sequential data where past information influences current decisions. In both resource allocation and routing, system conditions evolve rapidly, and LSTM's ability to adapt to new data enables effective decision-making as conditions change. The LSTM model includes the two LSTM blocks with a dense layer as shown in Fig. 3. The cell state is saved in the LSTM block, which may remove or add the data from the cell state. The gates are served as an optional data intake. Moreover, the point-wise multiplication and sigmoid layer are the major components of the LSTM. The forget gate (a_m) is formulated as,

$$a_m = \eta(K_m J^a + I_{m-1}\omega^a + I_a) \tag{1}$$

where η indicates sigmoid activation function, K_m denotes input vector, and I_{m-1} signifies a hidden state of previous layer.

Following that, the layer creates the input upgraded vector $j_m$value as:

$$j_m = \eta\left(K_m J^j + I_{m-1}\omega^j + I_j\right) \tag{2}$$

The vector of new candidate value $\widetilde{N}_m$ is formulated as,

$$\widetilde{N}_m = \tanh(K_m J^n + I_{m-1}\omega^{\bar{j}} + I_{\bar{j}}) \tag{3}$$

The previous cell state $N_{m-1}$is updated to the new cell state N_m as follows,

$$N_m = N_{m-1} \otimes a_m \oplus j_m \otimes \widetilde{N}_m \tag{4}$$

The output gate i_m detects which parts of the cell state yield the output.

$$i_m = \eta\left(K_m J^0 + I_{m-1}\omega^0 + I_0\right) \tag{5}$$

The hidden state I_m is expressed as,

$$I_m = i_m \otimes \tanh(N_m) \tag{6}$$

where, J^a, J^j, J^n, J^0 explicates input weights. Then, recurrent weights are indicated as ω^a, ω^j, $\omega^{\bar{j}}$ and ω^0. Bias terms are denoted as I_0, I_j, $I_{\bar{j}}$, and I_a.

From the output, the predicted energy$EN_{pr}$is obtained.

(b) Execution time

Execution time is described as the time consumed through the gateway for executing a task and it is expressed as,

$$T_{Exe} = \frac{1}{\max\left(T_{Exe}\right) * o} \sum_{o=1}^{d} \left(T_{Exe}^{Gate} * o\right) \tag{7}$$

The count of the task is indicated as$o$andT_{Exe}implies the execution time. Furthermore, the task performed in the gateway is portrayed byT_{Exe}^{Gate}.

(c) Communication cost

Communication cost is based on the whole resources present in the gateway, which is formulated below,

$$Z = \sum_{o=1}^{z_p} \varpi_{po} \tag{8}$$

where, ϖ_{po}implies the communication cost for the p^{th}gateway and the entire resources combined with it. Furthermore, $z_p$defines the count of joined resources for the gateway$o$.

4.1.2. Solution encoding

The solution encoding is designed to obtain the finite solution for RA. Here, the LJHGSO is used in RA, which effectively allocates the optimal resource to the gateway. Moreover, the solution encoding is given in

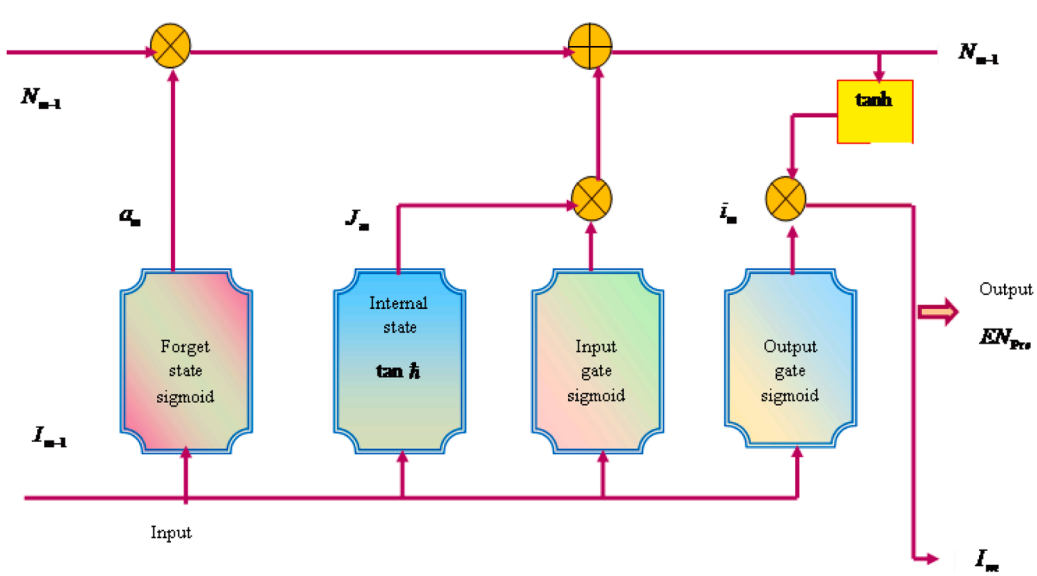

Fig. 3. Schematic illustration of deep LSTM.

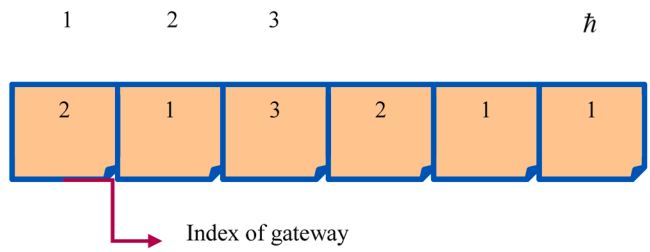

Fig. 4. Solution encoding.

Fig. 4.

4.1.3. Fitness function

The fitness function is utilized for detecting the ideal gateway to allocate the resources; hence, the energy effectual network is obtained. Furthermore, the execution time of the gateway, energy, and communication cost are the parameters to find the fitness.

$$Fitness = \frac{1}{3}\left[(1 - EN_{\mathrm{Pre}}) + Z + T_{Exe}\right] \tag{9}$$

where, EN_{Pre}implies the predicted energy, Zshows the communication cost, and T_{Exe}represents the execution time.

4.1.4. Algorithmic steps of LJHGS for optimal RA

The LJHGS is created by combining the LOA [19], Jaya algorithm [20], and HGS [21]. Here, the HGS is based on the attitude of starving animals, where the animal exhibits two primary behaviors such as cooperation and non-cooperation. It is utilized to balance exploration and exploitation. Moreover, the Jaya algorithm [20] is merged to resolve the unconstrained issues. Jaya Optimization is based on the concept of social interactions. Moreover, the Jaya algorithm modifies every solution via the attraction and improvement principles. Attraction entails moving toward the optimal solutions, and improvement entails the modification of the current solution to be near the ideal solution found so far. For attaining quick convergence, the LOA is added to Jaya-HGS. LOA is a kind of metaheuristic approach, which inspires the distinctive singing characteristics of lyrebirds. It is characterized by their capability to mimic different sounds from their surroundings. Moreover, the LOA offered the optimal solution even in complicated solution spaces. Integrating these three algorithms enhances system performance in terms of cost, longevity, and RA. The steps carried out in the LJHGS are described as follows,

Step 1: Initialization

InHGS [21], κcount of populations is presented in the search area, which is given as,

$$M = \{M_1, M_2, \ldots, M_\kappa\} \tag{10}$$

Step 2: Computation of fitness function

The fitness function provides the optimal solution, which is given in Eq. (9).

Step 3: Updated Equation for LJHGS

The mathematical form of non-cooperative behavior is given as,

$$M(\alpha+1) = \{A_1.M_c + B.A_2.|M_c - M(\alpha)|, \quad d_2 > \Re \tag{11}$$

Where$M(\alpha)$ shows the location of the present individual and$M_c$shows the best individual in the present iteration. Moreover, the terms $A_1$and A_2specifies the weight of the hungry. The random number is indicated as d_2,$\Re$exhibits the variable, which controls the global position, and the ranging controller is indicated by the term B.

Eq. (21) is written as,

$$M(\alpha+1) = A_1.M_c + B.A_2.M_c - B.A_2.M(\alpha) \tag{12}$$

$$M(\alpha+1) = M_c(A_1 + B.A_2) - B.A_2.M(\alpha) \tag{13}$$

In this phase, the Jaya algorithm [20] is added, and the updated solution of Jaya with HGSO is given as,

$$M(\alpha+1) = \frac{1}{[1-\beta_1+\beta_2+B.A_2]}\left[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2) - B.A_2(\beta_1 M_{best}(\alpha) - \beta_2 M_{worst}(\alpha))\right] \tag{14}$$

Here, $M_{best}$and M_{worst}shows the best candidate with the best and worst solution. Moreover, $\beta_1$and $\beta_2$represents the random number.

Here, the LOA [19] is added to the updated equation of Jaya with HGS [21], and the new location of the LOA is given as follows,

$$M_{e.f}^{g1} = M_{e.f} + \chi_{e.f}.\left(\gamma_{e.f} - H_{e.f}.M_{e.f}\right) \tag{15}$$

where, $\gamma_{e.f}$represents the selected safe area (SSA) for the e^{th}lyrebird with the dimension f, and $M_{e.f}^{g1}$represents the new location of the Lyrebird. Moreover, the random number γlies in [0,1] and the random number $H$is selected as s 1 or 2.

Consider, $M_{e.f}^{g1} = M(\alpha+1)$, $M_{e.f} = M(\alpha)$,$\chi_{e.f} = \chi$, $\gamma_{e.f} = \gamma$, and $H_{e.f} = H$

Hence, Eq. (15) becomes,

$$M(\alpha+1) = M(\alpha) + \chi.(\gamma - H.M(\alpha)) \tag{16}$$

$$M(\alpha+1) = M(\alpha)(1 - \chi.H) + \chi\gamma \tag{17}$$

$$M(\alpha) = \frac{M(\alpha+1) - \chi\gamma}{(1 - \chi.H)} \tag{18}$$

Subtract $M(\alpha)$ on both sides

$$M(\alpha+1) - M(\alpha) = \frac{1}{[1-\beta_1+\beta_2+B.A_2]}\left[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2) - B.A_2(\beta_1 M_{best}(\alpha) - \beta_2 M_{worst}(\alpha))\right] - M(\alpha) \tag{19}$$

Substitute Eq. (17) in Eq. (14)

$$M(\alpha+1)-M(\alpha) = \frac{1}{[1-\beta_1+\beta_2+B.A_2]}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))] - \frac{M(\alpha+1)-\chi\gamma}{(1-\chi.H)} \tag{20}$$

$$M(\alpha+1)+\frac{M(\alpha+1)}{(1-\chi.H)} = \frac{1}{[1-\beta_1+\beta_2+B.A_2]}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))] - \frac{\chi\gamma}{(1-\chi.H)}+M(\alpha) \tag{21}$$

$$\frac{M(\alpha+1)(1-\chi.H)+M(\alpha+1)}{(1-\chi.H)} = \frac{1}{[1-\beta_1+\beta_2+B.A_2]}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))] - \frac{\chi\gamma}{(1-\chi.H)}+M(\alpha) \tag{22}$$

$$\frac{M(\alpha+1)(1-\chi.H)+M(\alpha+1)}{(1-\chi.H)} = \frac{\left[\begin{array}{l}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))](1-\chi.H)- \\ \chi\gamma[1-\beta_1+\beta_2+B.A_2] \ +M(\alpha)(1-\chi.H)[1-\beta_1+\beta_2+B.A_2]\end{array}\right]}{(1-\chi.H)[1-\beta_1+\beta_2+B.A_2]} \tag{23}$$

$$M(\alpha+1)(1-\chi.H)+M(\alpha+1) = \frac{\begin{array}{c}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))] \\ (1-\chi.H)-\chi\gamma[1-\beta_1+\beta_2+B.A_2] \ +M(\alpha)(1-\chi.H)[1-\beta_1+\beta_2+B.A_2]\end{array}}{1-\beta_1+\beta_2+B.A_2} \tag{24}$$

$$M(\alpha+1)(2-\chi.H) = \frac{\begin{array}{r}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))](1-\chi.H)- \\ \chi\gamma[1-\beta_1+\beta_2+B.A_2] \ +M(\alpha)(1-\chi.H)[1-\beta_1+\beta_2+B.A_2]\end{array}}{1-\beta_1+\beta_2+B.A_2} \tag{25}$$

$$M(\alpha+1)=\frac{\begin{array}{c}[(1-\beta_1+\beta_2)M_c.(A_1+B.A_2)-B.A_2(\beta_1 M_{best}(\alpha)-\beta_2 M_{worst}(\alpha))](1-\chi.H)-\\ \chi\gamma[1-\beta_1+\beta_2+B.A_2]\ +M(\alpha)(1-\chi.H)[1-\beta_1+\beta_2+B.A_2]\end{array}}{(1-\beta_1+\beta_2+B.A_2)(2-\chi.H)} \tag{26}$$

Step 4: Termination

The steps are continued until the ideal solution is attained.

4.2. Congestion-aware routing using proposed LFWBO

The congestion-aware routing in IoT is used to offer a superior path to the gateway without congestion. Moreover, the routing is carried out by considering the fitness parameters like queue length, link quality, energy, and communication cost.

4.2.1. Solution encoding

Fig. 5 depicts the solution encoding for routing. Here, the parent hub of the network is selected as the CH. Here, the LFWBO is used for performing the effectual routing.

4.2.2. Fitness function

Fitness is determined to attain the superior route with the congestion. The expression for fitness is given below,

$$Fitness=\frac{1}{4}[LQ+QL+(1-Dis)+EN_{\text{Pre}}] \tag{27}$$

Here, LQrepresents the link quality, QLspecifies the Queue length,Dis portrays the distance, and EN_{Pre}exhibits the predicted energy.

(a) Queue Length

QL [22] is the fitness parameter in the routing process, which considers the congestion for every node. Moreover, the QL is employed to improve the rate of data delivery.

$$QL=\frac{D_q}{C} \tag{28}$$

Here, $D_q$implies the received packets on the node$q$, and the total buffers are indicated as C.

(b) Link Quality

Link quality [23] is employed for identifying the data delivery among the nodes, which is based on the count of transmissions and retransmissions of the data packet. Moreover, the link quality is expressed as,

$$LQ=\frac{1}{r*s} \tag{29}$$

Here,$r$and$s$represents the forward and reverse transmission of the data.

(c) Distance

Distance represents the measure of space between two nodes within a network. The following expression is used to specify the distance.

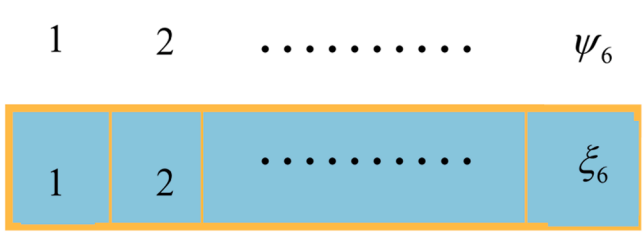

Fig. 5. Block diagram of solution encoding.

$$Dis_{no_1,no_2}\rightarrow \parallel Dis_{no_1}-Dis_{no_2}\parallel \tag{30}$$

where, Dis_{no_1,no_2}represents the distance among the nodes no_1, and no_2.

(d) Predicted Energy

The predicted energy is considered one of the crucial fitness parameters, which is already explained in Section 4.1.1.

4.2.3. Proposed LFWBO-based congestion-aware routing

In congestion-aware routing, the LFWBO is employed. Here, the merging of LOA [18]as well as FWBO creates the LFWBO. The peculiar singing characteristics of the lyrebirds are the inspiration of LOA. It provided the optimal solution for complex search spaces. Moreover, the FWBO is added to LOA to obtain a superior solution. WOA [23] is a nature-inspired algorithm, which is based on the foraging habits, and social behavior of the walrus. The optimal exploration and exploitation of WOA enable faster convergence. Furthermore, DBO [24] is based on the distinctive attitude of dung beetles, especially their foraging and navigation strategies. Dung beetles are the widespread insects that feed on animal dung. Moreover, the DBO yields the appropriate solution for a variety of optimization issues in discrete and combinatorial tasks. FC [25] is a mathematical analysis, that includes the integration and differentiation in non-integer orders for providing a reliable solution. The algorithmic steps for LFWBO are explained below:

Step 1: Initialization

The population members are initialized randomly, which is given as,

$$L=\{L_1,L_2,.....L_b,.....L_k\} \tag{31}$$

Here, the population members are denoted as L, the entire members are denoted as k, and the b^{th}member is denoted as L_b.

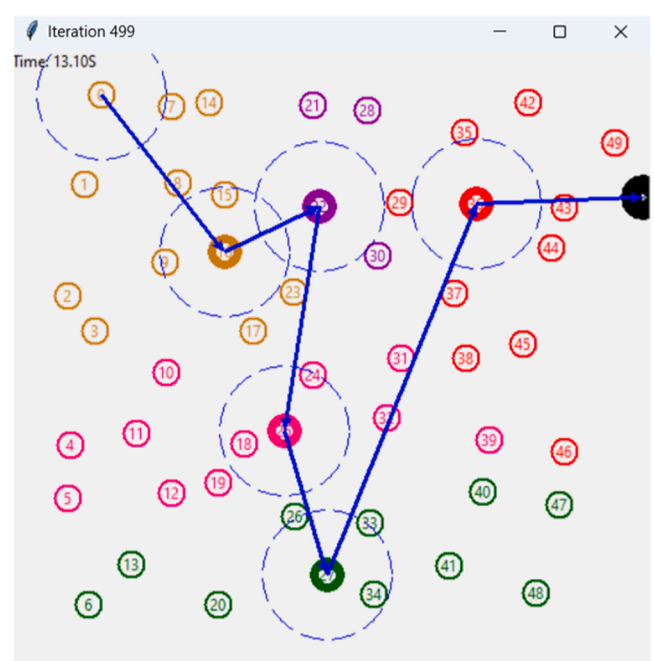

Fig. 6. Simulation result for routing.

Step 2: Fitness Function

Fitness is required to obtain the ideal solution, and the expression for fitness is given in Eq. (27)

Step 3: Updated solution for LFWBO

The LOA [19] is added to the updated solution of FWBO for obtaining the final solution of LFWBO. Here, the updated solution of WBO $(L_b(\lambda+1))$ is given as follows,

$$L_b(\lambda+1) = \frac{1}{rand \cdot G - \tan(\theta)}[rand\,\delta(1+\tan(\theta)) + \tan(\theta)L_b(\lambda-1)(1-rand \cdot G)] + \varepsilon L_b(\lambda) + \frac{1}{2}\varepsilon L_b(\lambda-1) + \frac{1}{6}(1-\varepsilon)L_b(\lambda-2) + \frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3) \tag{32}$$

where, θ implies the deflection angle $(0, \pi)$,δindicates the strongest walrus with the best value, and $_G$ represents the integer selected randomly (1 or 2). The term *rand*represents the random number, εshows the constant. Moreover, the updated position of LOA $(L(\lambda+1))$ is expressed as,

$$L(\lambda+1) = L(\lambda) + \chi.(\gamma - H.L(\lambda)) \tag{33}$$

where, γshows the SSA for the e^{th}lyrebird with the dimension f, the random number $H$is selected as s 1 or 2. Moreover, the random number γlies in $[0, 1]$ interval.

Consider, $L(\lambda+1) = L_b(\lambda+1)$, $L(\lambda) - L_b(\lambda)$,$\chi = \chi_b$, and $H = H_b$ Hence, Eq. (33) becomes,

$$L_b(\lambda+1) = L_b(\lambda) + \chi_b.(\gamma - H_b.L_b(\lambda)) \tag{34}$$

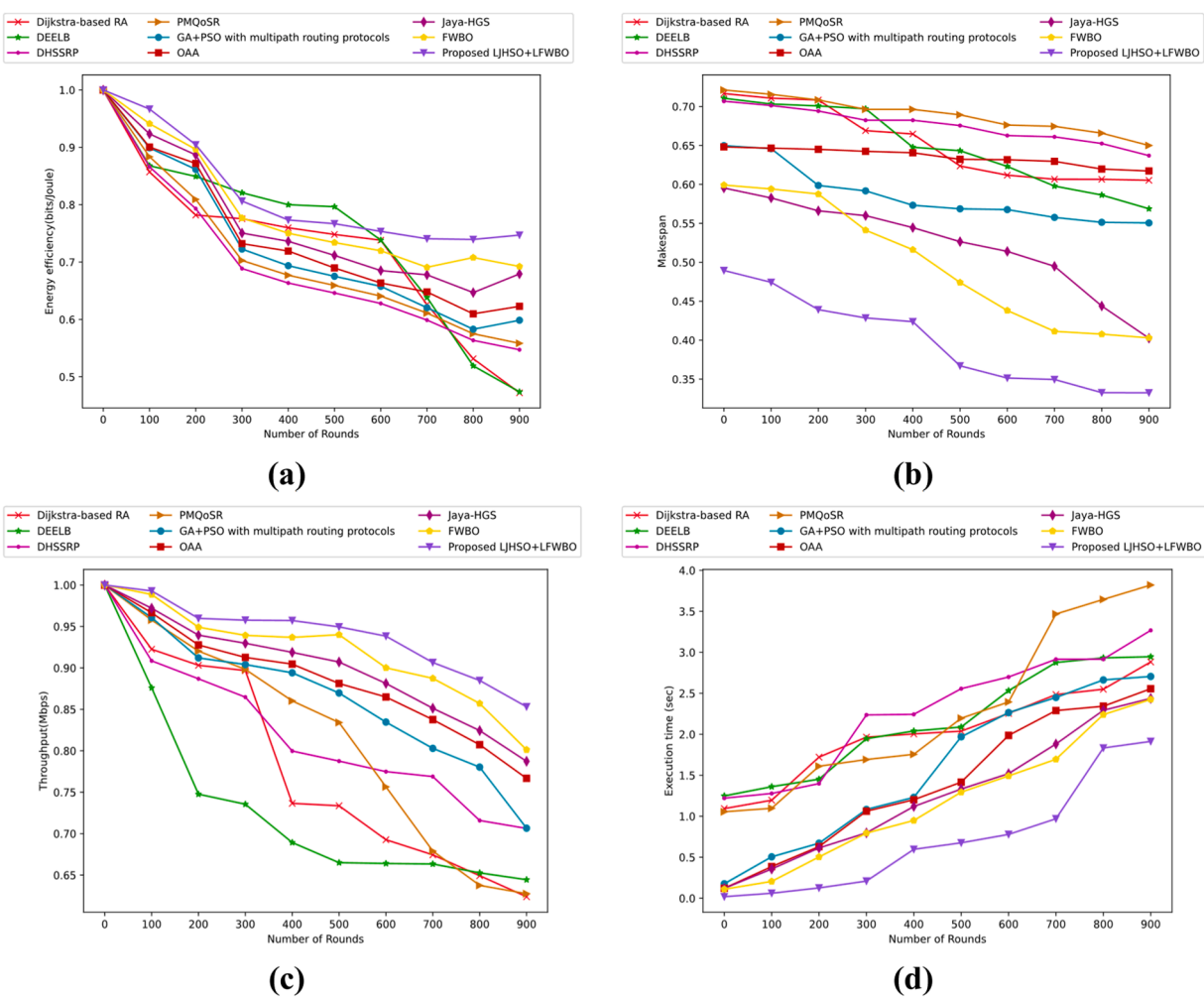

Fig. 7. Comparative analysis of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT with 50 nodes, a) Energy b) Makespan, c) Throughput, d) Execution time.

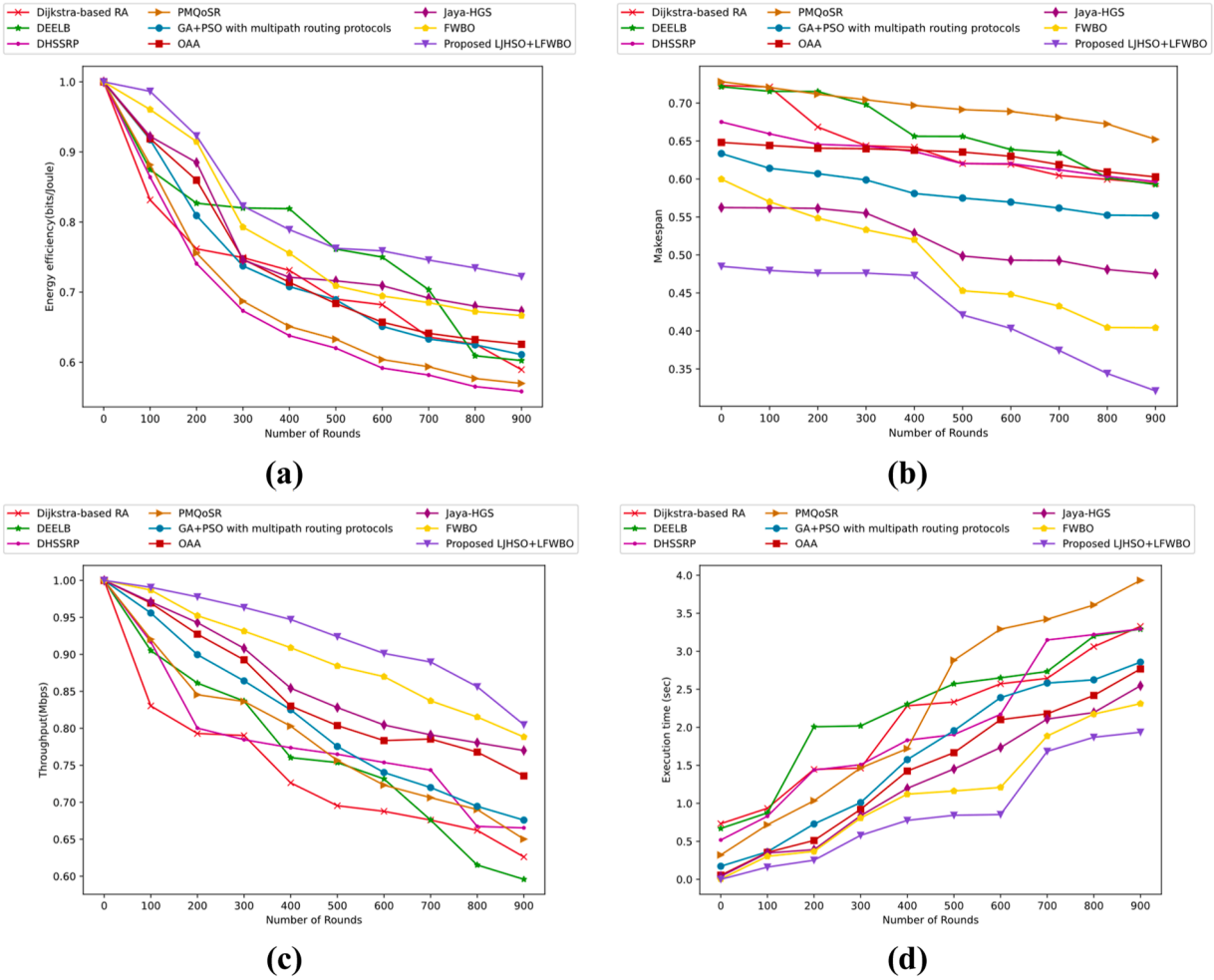

Fig. 8. Comparative analysis of LJHSO+LFWBO-based joint RA and congestion aware routing in IoT with 100 nodes, (a) Energy (b) Makespan, (c) Throughput, (d) Execution time.

$$L_b(\lambda+1) = L_b(\lambda)(1-\chi_b.H_b) + \chi_b\gamma \tag{35}$$

$$L_b(\lambda) = \frac{L_b(\lambda+1) - \chi_b\gamma}{(1-\chi_b.H_b)} \tag{36}$$

$$\begin{aligned} L_b(\lambda+1) = {} & \frac{1}{rand\cdot G - \tan(\theta)}[rand\,\delta(1+\tan(\theta)) + \tan(\theta)L_b(\lambda-1)(1-rand\cdot G)] + \\ & \varepsilon\left(\frac{L_b(\lambda+1) - \chi_b\gamma}{(1-\chi_b.H_b)}\right) + \frac{1}{2}\varepsilon L_b(\lambda-1) + \frac{1}{6}(1-\varepsilon)L_b(\lambda-2) + \frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3) \end{aligned} \tag{37}$$

$$\begin{aligned} L_b(\lambda+1) - \varepsilon\left(\frac{L_b(\lambda+1)}{(1-\chi_b.H_b)}\right) = {} & \frac{1}{rand\cdot G - \tan(\theta)}[rand\,\delta(1+\tan(\theta)) + \tan(\theta)L_b(\lambda-1)(1-rand\cdot G)] - \\ & \varepsilon\frac{\chi_b\gamma}{(1-\chi_b.H_b)} + \frac{1}{2}\varepsilon L_b(\lambda-1) + \frac{1}{6}(1-\varepsilon)L_b(\lambda-2) + \frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3) \end{aligned} \tag{38}$$

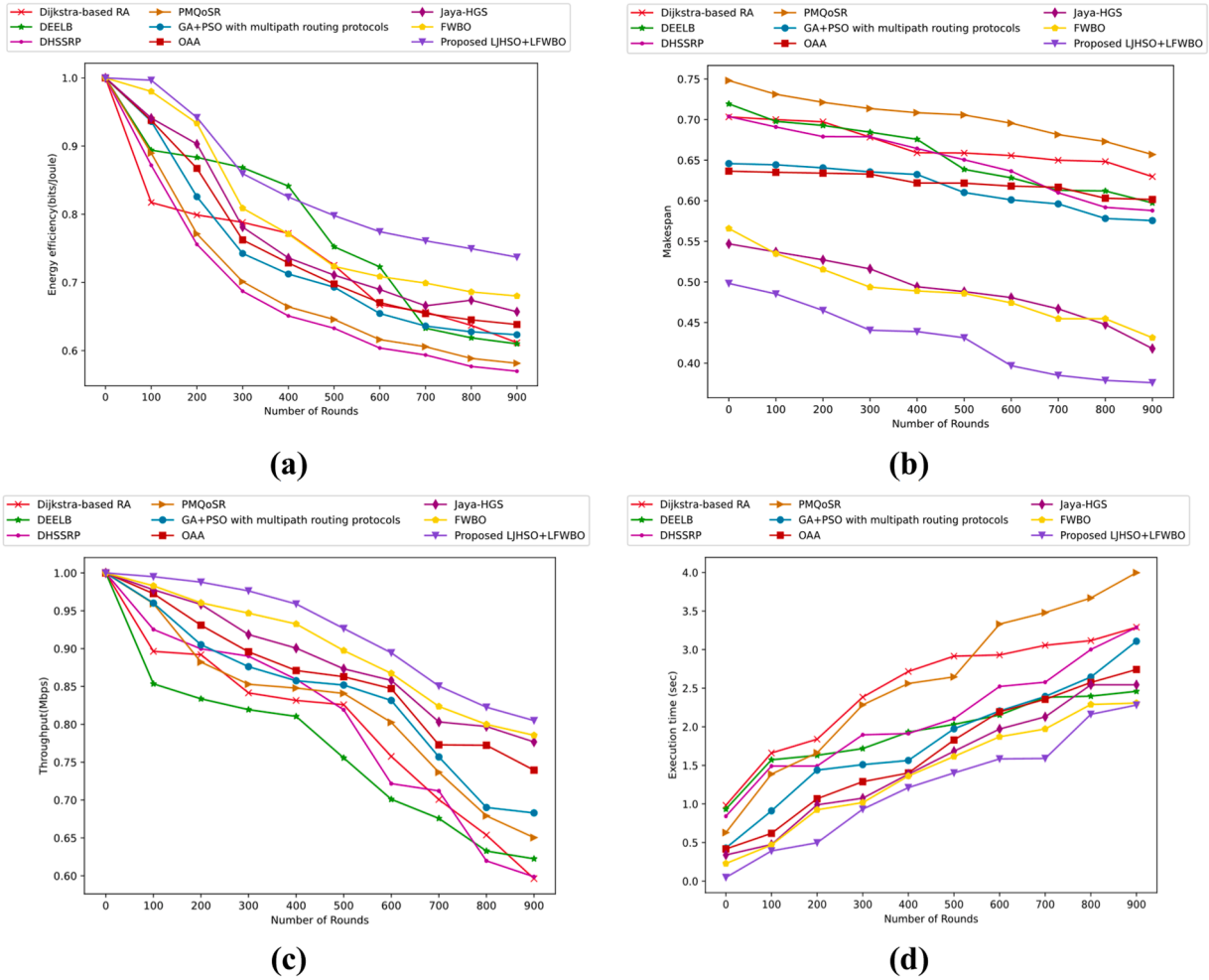

Fig. 9. Comparative analysis of LJHSO+LFWBO-based joint RA and congestion aware routing in IoT with 150 nodes, a) Energy b) Makespan, c) Throughput, d) Execution time.

Table 2
Comparative discussion.

Variations	Metrics/Methods	Dijkstra-based RA	DEELB	DHSSRP	PMQoSR	GA+PSO with multipath routing protocols	OAA	Jaya-HGS	FWBO	Proposed LJHSO+LFWBO
50 nodes	***Energy (bits/Joule)***	0.472	0.474	0.547	0.558	0.599	0.623	0.679	0.692	**0.747**
	Makespan	0.605	0.569	0.637	0.650	0.551	0.617	0.403	0.403	**0.332**
	Throughput (Mbps)	0.624	0.644	0.706	0.628	0.707	0.767	0.787	0.801	**0.853**
	Execution time (sec)	2.883	2.947	3.268	3.821	2.706	2.556	2.438	2.426	**1.913**
100 nodes	***Energy (bits/Joule)***	0.589	0.602	0.558	0.570	0.611	0.626	0.673	0.666	0.722
	Makespan	0.594	0.593	0.597	0.652	0.552	0.603	0.475	0.404	0.321
	Throughput (Mbps)	0.626	0.596	0.665	0.650	0.676	0.736	0.770	0.788	0.805
	Execution time (sec)	3.327	3.291	3.293	3.932	2.857	2.767	2.546	2.311	1.936
150 nodes	***Energy (bits/Joule)***	0.612	0.610	0.570	0.581	0.623	0.638	0.657	0.680	0.737
	Makespan	0.630	0.597	0.588	0.657	0.576	0.602	0.418	0.431	0.376
	Throughput (Mbps)	0.596	0.622	0.599	0.650	0.683	0.739	0.777	0.786	0.805
	Execution time (sec)	3.29	2.46	3.288	3.997	3.108	2.743	2.544	2.308	2.281

$$\frac{L_b(\lambda+1)(1-\chi_b.H_b)-\varepsilon L_b(\lambda+1)}{(1-\chi_b.H_b)}=\frac{\begin{array}{c}[rand\ \delta(1+\tan(\theta))+\tan(\theta)L_b(\lambda-1)(1-rand\cdot G)](1-\chi_b.H_b)-\varepsilon\chi_b\gamma(rand\cdot G-\tan(\theta))\ +\\ \left[\frac{1}{2}\varepsilon L_b(\lambda-1)+\frac{1}{6}(1-\varepsilon)L_b(\lambda-2)+\frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3)\right](1-\chi_b.H_b)(rand\cdot G-\tan(\theta))\end{array}}{rand\cdot G-\tan(\theta)(1-\chi_b.H_b)} \tag{39}$$

$$\frac{L_b(\lambda+1)[1-\chi_b.H_b-\varepsilon]}{(1-\chi_b.H_b)}=\frac{\begin{array}{c}[rand\ \delta(1+\tan(\theta))+\tan(\theta)L_b(\lambda-1)(1-rand\cdot G)](1-\chi_b.H_b)-\varepsilon\chi_b\gamma(rand\cdot G-\tan(\theta))\ +\\ \left[\frac{1}{2}\varepsilon L_b(\lambda-1)+\frac{1}{6}(1-\varepsilon)L_b(\lambda-2)+\frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3)\right](1-\chi_b.H_b)(rand\cdot G-\tan(\theta))\end{array}}{rand\cdot G-\tan(\theta)(1-\chi_b.H_b)} \tag{40}$$

$$L_b(\lambda+1)[1-\chi_b.H_b-\varepsilon]=\frac{\begin{array}{c}[rand\ \delta(1+\tan(\theta))+\tan(\theta)L_b(\lambda-1)(1-rand\cdot G)](1-\chi_b.H_b)-\varepsilon\chi_b\gamma(rand\cdot G-\tan(\theta))\ +\\ \left[\frac{1}{2}\varepsilon L_b(\lambda-1)+\frac{1}{6}(1-\varepsilon)L_b(\lambda-2)+\frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3)\right](1-\chi_b.H_b)(rand\cdot G-\tan(\theta))\end{array}}{rand\cdot G-\tan(\theta)} \tag{41}$$

$$L_b(\lambda+1)=\frac{\begin{array}{c}[rand\ \delta(1+\tan(\theta))+\tan(\theta)L_b(\lambda-1)(1-rand\cdot G)](1-\chi_b.H_b)-\varepsilon\chi_b\gamma(rand\cdot G-\tan(\theta))\ +\\ \left[\frac{1}{2}\varepsilon L_b(\lambda-1)+\frac{1}{6}(1-\varepsilon)L_b(\lambda-2)+\frac{1}{24}\varepsilon(1-\varepsilon)(2-\varepsilon)L_b(\lambda-3)\right](1-\chi_b.H_b)(rand\cdot G-\tan(\theta))\end{array}}{rand\cdot G-\tan(\theta)[1-\chi_b.H_b-\varepsilon]} \tag{42}$$

Step 4: Termination

The steps are continued until the ideal solution is attained.

5. Results and discussion

In this section, the experimental assessment of proposed LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT is described. Moreover, the evaluation parameters, implementing tool, and comparative evaluation are also illustrated.

5.1. Experimental setup

LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT are implemented in the Python tool. The system requires Windows 10 as the operating system, 8GB of RAM, and >100GB of storage (ROM), with a CPU clock speed of at least 1.7 GHz. A dedicated GPU is not needed for the setup. For software, Python version 3.9.11 is required, and it can be downloaded from the official Python website via the Windows installer (64-bit). Additionally, the system must have several specific libraries installed, including keras 2.13.1, tensorboard 2.13.0, wsnsimpy 0.2.5, Scipy 1.10.1, Simpy 3.0.11, Pillow 8.1.2, protobuf 4.24.3, numpy 1.23.5, and fuzzy-c-means 0.0.6.

5.2. Simulation result

The simulation outputs of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT are shown in Fig. 6. Here, the dark color nodes in orange, green, purple, and pink color represent the CH, the black color indicates the BS, and the remaining nodes specify the cluster members.

5.3. Performance metrics

The performance of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT is analyzed using the energy, makespan, throughput, and execution time.

5.3.1. Energy

The energy parameter is crucial to provide effectual routing in IoT, which is described in Section 4.1.1.

5.3.2. Makespan

The time gap between the first processing unit initiating the task and the completion of the task in every processing unit describes the Makespan [26].

Table 3
ANOVA test.

	Sum of squares	*Degrees of freedom*	*F*	*P-Value*
C (treatments)	0.179656	4	15.675789	0.00049
Residual	0.216546	120		

5.3.3. Throughput

The throughput [27] is described as the counting of data packets transmitted at a certain time, which is given by

$$Throughput = \frac{\mathbb{k}_{re}}{\hbar} \tag{43}$$

where, $\mathbb{k}_{re}$implies the received packet at the time $\hbar$.

5.3.4. Execution time

The execution time is denoted as the time needed to execute the data transmission, which is given in Eq. (7).

5.4. Comparative methods

The methods like Dijkstra-based RA [8], DEELB [13], DHSSRP [9], PMQoSR [10], GA+PSO with multipath routing protocols [30], OAA [31], Jaya-HGS, and FWBO are considered for validating the performance of LJHSO+LFWBO.

5.5. Comparative analysis

The comparative estimation of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT is carried out by considering different nodes like 50, 100, and 150.

5.5.1. Analysis using 50 nodes

The comparative analysis of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT with 50 nodes is shown in Fig. 7. The analysis is performed by considering the energy, makespan, throughput, and computation time. The analysis in terms of energy is portrayed in Fig. 7a. Using the 900 rounds, the energy of LJHSO+LFWBO is 0.747 bits/Joule, in which the existing models, like Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO attained the energy of 0.472 bits/Joule, 0.474bits/Joule, 0.547bits/Joule, 0.558bits/Joule, 0.599bits/Joule, 0.623bits/Joule, 0.679bits/Joule, and 0.692bits/Joule. The analysis concerning makespan is shown in Fig. 7b. For 900 rounds, the makespan of LJHSO+LFWBO is 0.332, in which the makespan of 0.605, 0.569, 0.637, 0.650, 0.551, 0.617, 0.403, and 0.403. are obtained by the Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO. Fig. 7c portrays the assessment regarding throughput. Here, the LJHSO+LFWBO attained the throughput of 0.853 Mbps, in which the throughput of Dijkstra-based RA is 0.624 Mbps, DEELB is 0.644 Mbps, DHSSRP is 0.706 Mbps, PMQoSR is 0.628 Mbps, GA+PSO with multipath routing protocols is 0.707 Mbps, OAA is 0.767 Mbps, Jaya-HGS is 0.787 Mbps, and FWBO is 0.801 Mbps under the 900th round. Moreover, the evaluation concerning execution time is illustrated in Fig. 7d. For the 900th round, the LJHSO+LFWBO attained the execution time of 1.913 s, whereas the existing models got execution time of 2.883 s, 2.947 s, 3.268 s, 3.821 s, 2.706 s, 2.556 s, 2.438 s, and 2.426 s.

5.5.2. Analysis using 100 nodes

Fig. 8 deliberates the estimation concerning 100 nodes. The analysis regarding energy is illustrated in Fig. 8a. For 900 rounds, the energy of LJHSO+LFWBO is 0.722bits/Joule, whereas the energy of 0.589bits/Joule, 0.602bits/Joule, 0.558bits/Joule, 0.570bits/Joule, 0.611bits/Joule, 0.626bits/Joule, 0.673bits/Joule, and 0.666bits/Joule are attained by the Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO. The assessment regarding makespan is displayed in Fig. 8b. In 900 rounds, the makespan of LJHSO+LFWBO is 0.321, whereas the makespan of Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO are 0.594, 0.593, 0.597, 0.652, 0.552, 0.603, 0.475, and 0.404. Fig. 8c exhibits the analysis regarding throughput. For 900 rounds, the LJHSO+LFWBO achieved the throughput of 0.805 Mbps, whereas the throughput of Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO are 0.626 Mbps, 0.596 Mbps, 0.665 Mbps, 0.650 Mbps, 0.676 Mbps, 0.736 Mbps, 0.770 Mbps, and 0.788 Mbps. The assessment in terms of execution time is given in Fig. 8d. Using 900 rounds, the LJHSO+LFWBO attained the execution time of 1.936 s, whereas the execution time of Dijkstra-based RA is 3.327 s, DEELB is 3.291 s, DHSSRP is 3.293 s, PMQoSR is 3.932, GA+PSO with multipath routing protocols is 2.857 s, OAA is 2.767 s, Jaya-HGS is 2.546 s, and FWBO is 2.311 s.

5.5.3. Analysis using 150 nodes

Fig. 9 exhibits the comparative estimation of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT using 150 nodes. The estimation concerning energy is illustrated in Fig. 9a. For 900 rounds, the energy of LJHSO+LFWBO is 0.737bits/Joule, whereas the energy of Dijkstra-based RA is 0.612bits/Joule, DEELB is 0.610 bits/Joule, DHSSRP is 0.570 bits/Joule, PMQoSR is 0.581 bits/Joule, GA+PSO with multipath routing protocols is 0.623 bits/Joule, OAA is 0.638 bits/Joule, Jaya-HGS is 0.657 bits/Joule, and FWBO is 0.680 bits/Joule. The estimation in terms of makespan is portrayed in Fig. 9b. While considering 900 rounds, the makespan of LJHSO+LFWBO is 0.376, whereas the makespan of Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO are 0.630, 0.597, 0.588, 0.657, 0.576, 0.602, 0.418, and 0.431. Fig. 9c displays the assessment concerning throughput. Here, the throughput of LJHSO+LFWBO is 0.805 Mbps, in which the throughput of Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO are 0.596 Mbps, 0.622 Mbps, 0.599 Mbps, 0.650 Mbps, 0.683 Mbps, 0.739 Mbps, 0.777 Mbps, and 0.786 Mbps under the 900 round. In addition, the evaluation regarding execution time is deliberated in Fig. 9. For 900 rounds, the execution time of 3.290 s, 2.460 s, 3.288 s, 3.997 s, 3.108 s, 2.743 s, 2.544 s, 2.308 s, and 2.281 s are attained by the Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, FWBO, and LJHSO+LFWBO.

5.6. Comparative discussion

The comparative discussion of LJHSO+LFWBO-based joint RA and congestion-aware routing in IoT is presented in Table 2. Here, the metrics like energy, makespan, throughput, and execution time are considered for assessing the performance of the model, in which the highest values of energy, throughput, and the minimum value of makespan, and execution time are considered as the optimal values. The highest energy rate achieved by the proposed LJHSO+LFWBO is 0.747 bits/Joule, while prior models, like Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO obtained 0.472 bits/Joule, 0.474 bits/Joule, 0.547 bits/Joule, 0.558 bits/Joule, 0.599 bits/Joule, 0.623 bits/Joule, 0.679 bits/Joule, and 0.692 bits/Joule for 50 nodes. The superior makespan rate gained by LJHSO+LFWBO is 0.332, while existing approaches, like Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO attained 0.605, 0.569, 0.637, 0.650, 0.551, 0.617, 0.403, and 0.403 for 50 nodes. For 50 nodes, throughput value acquired by proposed LJHSO+LFWBO is 0.853, while Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO reached 0.624 Mbps,

0.644 Mbps, 0.706 Mbps, 0.628 Mbps, 0.707 Mbps, 0.767 Mbps, 0.787 Mbps, and 0.801 Mbps. For 50 nodes, execution time reached by proposed LJHSO+LFWBO, Dijkstra-based RA, DEELB, DHSSRP, PMQoSR, GA+PSO with multipath routing protocols, OAA, Jaya-HGS, and FWBO are 1.913 s, 2.883 s, 2.947 s, 3.268 s, 3.821 s, 2.706 s, 2.556 s, 2.438 s, and 2.426 s. The hybridization of multiple optimization techniques (Lyrebird, Jaya, and Hunger Search) enhances the algorithm's ability to converge more quickly and accurately toward the optimal solution. LJHSO is suitable for large-scale optimization problems, as it is designed to efficiently handle a significant number of variables and constraints, which is especially important in real-world applications like resource management in IoT networks. The **Lyrebird** component suggests that the algorithm could imitate the most successful strategies from multiple sources, which would help it adaptively explore different regions of the solution space (exploration). The **Fractional** aspect allows the algorithm to carefully refine and focus on areas that have shown potential (exploitation). This dynamic balance makes **LFWBO** effective in finding optimal solutions without getting trapped in local optima. Walrus Optimization has the collective search behavior. By considering the strengths of this above-mentioned optimization that are combined to generate effective RA and congestion-aware routing. Thus, a remarkable novelty is created by combining these optimizations.

5.6.1. Statistical analysis

The statistical analysis is often exploited by researchers to analyze the pattern. Complex data can be made **easier to understand** by representing it in tabular form. Instead of dealing with raw numbers, statistical tables distill the most important information and display it in an accessible format. ANOVA (Analysis of Variance) is a statistical test used to determine if there are significant differences between the means of three or more groups. It compares the variance between group means to the variance within the groups to assess whether group differences are due to chance. ANOVA is commonly used in experimental studies with multiple groups or conditions. The test results include an F-statistic and a P-value to evaluate statistical significance. Key assumptions include independence, normality, and equal variances across groups. Statistics of the proposed approach are computed by ANOVA test and the results of the ANOVA test are illustrated in Table 3.

5.6.2. Practical applicability

The newly developed LJHSO+LFWBO is applicable in real-time as it generated less execution time, high throughput, high energy, and low makespan time when compared to existing systems. LJHSO+LFWBO will be employed to optimize investment portfolios, balancing risk and return by considering the change in market conditions effectively. Moreover, Intelligent Transportation Systems (ITS) will be strengthened. LJHSO+LFWBO will be applied to multi-objective optimization problems where multiple conflicting goals need to be optimized simultaneously.

5.7. Limitations of present study

The limitations of the proposed model are described as follows:

LJHSO for RA faced scalability issues in large-scale systems, as it might struggle to efficiently handle dynamic changes in resource availability. Additionally, its ability to balance exploration and exploitation in complex scenarios could hinder optimal solutions. LJHSO encountered challenges in managing constraints and was sensitive to its parameter settings, reducing its adaptability. For **LFWBO** in congestion-aware routing for IoT, real-time decision-making in dynamic environments and handling IoT's inherent uncertainty are major hurdles. The algorithm might also struggle with high dimensionality, energy trade-offs, and the complexity of multi-metric routing in heterogeneous networks.

6. Conclusion

The rapid development of IoT devices offered an extraordinary count of connected devices.RA and congestion control are crucial aspects of controlling the interconnected devices in the IoT. Moreover, the effectual utilization of resources and congestion management are essential to provide the superior performance and reliability of IoT. In this work, the LJHSO-based RA and LFWBO-based congestion-aware routing is proposed. The IoT simulation is the beginning process, as well as the joint RA and routing, are performed after the simulation. The sensing, network, data processing, and application layers are part of IoT. The proposed LJHSO is employed for RA, which is done in the physical or sensing layer. Furthermore, the LFWBO with fitness parameters like queue length, distance energy, and link quality are used to perform the congestion-aware routing in the network layer. In addition, the energy, makespan, throughput, and execution time metrics are used to validate the effectiveness of the LJHSO+LFWBO-based joint RA and congestion-aware routing, in which the optimal values 0.747, 0.332, 0.853 Mbps, and 1.913 mins are achieved. The future direction of this model would hybridize with well-known metaheuristic algorithms such as **GA, PSO**, or **Differential Evolution (DE)** to enhance their exploration and exploitation abilities. Additionally, LJHSO+LFWBO will explore **Human-In-The-Loop (HITL) optimization,** where human input or feedback is integrated into the optimization process. Furthermore, it will focus on extending the optimization algorithms to address challenges posed by mobile terminals in the IoT. This includes handling dynamic topologies, mobility, and varying network conditions, as well as the potential integration with 5 G and edge computing networks.

CRediT authorship contribution statement

Yannam Bharath Bhushan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing. **S. Aparna:** Supervision.

Declaration of competing interest

The authors declare no conflict of interest

Data availability

No new data were generated or analyzed in this study.

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